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Program:

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GNU nano 8.7.1 skill_4.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 200
#define MAX_TOKENS 50

typedef struct {
    char *value;
} Token;

typedef struct {
    Token tokens[MAX_TOKENS];
    int count;
} TokenList;

// Check whether character is a delimiter
int is_delimiter(char c) {
    return c == '|' || c == '>' || c == '<' || c == ';';
}

// Tokenize input
void tokenize(char *input, TokenList *list) {
    list->count = 0;

    char *token = strtok(input, " \\t\\n");

    while (token != NULL && list->count < MAX_TOKENS) {
        list->tokens[list->count].value = malloc(strlen(token) + 1);

        if (list->tokens[list->count].value == NULL) {
            printf("Memory allocation failed\\n");
            exit(1);
        }
    }
}
```

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        strcpy(list->tokens[list->count].value, token);

        list->count++;

        token = strtok(NULL, " \\t\\n");
    }
}

// Display tokens
void display_tokens(TokenList *list) {

    printf("\nTokens:\n");

    for (int i = 0; i < list->count; i++) {
        printf("Token %d: %s\n",
            i + 1,
            list->tokens[i].value);
    }
}

// Validate token stream
int validate_tokens(TokenList *list) {

    if (list->count == 0) {
        printf("Empty command\n");
        return 0;
    }

    // Command cannot start with pipe
    if (strcmp(list->tokens[0].value, "|") == 0) {
        printf("Syntax Error: command cannot start with '|'\\n");
        return 0;
    }

    // Command cannot end with pipe
    if (strcmp(list->tokens[list->count - 1].value, "|") == 0) {
        printf("Syntax Error: command cannot end with '|'\\n");
        return 0;
    }
}
```

```

// Two consecutive pipes are invalid
for (int i = 0; i < list->count - 1; i++) {

    if (strcmp(list->tokens[i].value, "|") == 0 &&
        strcmp(list->tokens[i + 1].value, "|") == 0) {

        printf("Syntax Error: consecutive pipes\n");
        return 0;
    }
}

return 1;
}

// Simple parse tree representation
void create_parse_tree(TokenList *list) {

    printf("\nParse Tree:\n");

    printf("COMMAND\n");

    for (int i = 0; i < list->count; i++) {

        if (strcmp(list->tokens[i].value, "|") == 0) {
            printf(" PIPE\n");
        }
        else if (strcmp(list->tokens[i].value, ">") == 0) {
            printf(" OUTPUT_REDIRECTION\n");
        }
        else if (strcmp(list->tokens[i].value, "<") == 0) {
            printf(" INPUT_REDIRECTION\n");
        }
        else {
            printf(" ARGUMENT: %s\n",
                list->tokens[i].value);
        }
    }
}
}

```

```

int main() {
    char input[MAX_INPUT];

    TokenList list;

    printf("Enter a command: ");

    fgets(input, sizeof(input), stdin);

    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    // Handle empty command
    if (strlen(input) == 0) {
        printf("Empty command\n");
        return 0;
    }

    tokenize(input, &list);

    display_tokens(&list);

    if (validate_tokens(&list)) {
        printf("\nSyntax is valid.\n");

        create_parse_tree(&list);

        printf("\nExecution structure created successfully.\n");
    }
    else {
        printf("\nInvalid command.\n");
    }

    free_tokens(&list);

    return 0;
}

```

Output:

```
sahasrasagi@Sahasra:~/OS_Lab/skill$ ./skill_4
Enter a command: ls -l | grep txt

Tokens:
Token 1: ls
Token 2: -l
Token 3: |
Token 4: grep
Token 5: txt

Syntax is valid.

Parse Tree:
COMMAND
  ARGUMENT: ls
  ARGUMENT: -l
  PIPE
  ARGUMENT: grep
  ARGUMENT: txt

Execution structure created successfully.
sahasrasagi@Sahasra:~/OS_Lab/skill$ ./skill_4
Enter a command: ls || grep

Tokens:
Token 1: ls
Token 2: ||
Token 3: grep

Syntax is valid.

Parse Tree:
COMMAND
  ARGUMENT: ls
  ARGUMENT: ||
  ARGUMENT: grep

Execution structure created successfully.
sahasrasagi@Sahasra:~/OS_Lab/skill$ |
```